KANSER Paneli: Refinement Raporu

NB29 | 2026-06-22

Yonetici Ozeti

KANSER (hereditary cancer) panelinin %80/20 F1 skorunu yukseltmek icin 6 deney (E0-E5) karsilastirildi. En iyi: E1_COMBINED (LOO-MCC(prior)=0.6542, Boot-mean %80/20 F1=0.7143 +/- 0.0412, AUPRC=0.9643). NB16 referans: stack_lr %80/20 F1=0.716. Fark: -0.0017 -> NB16 referansi gecilemedi.

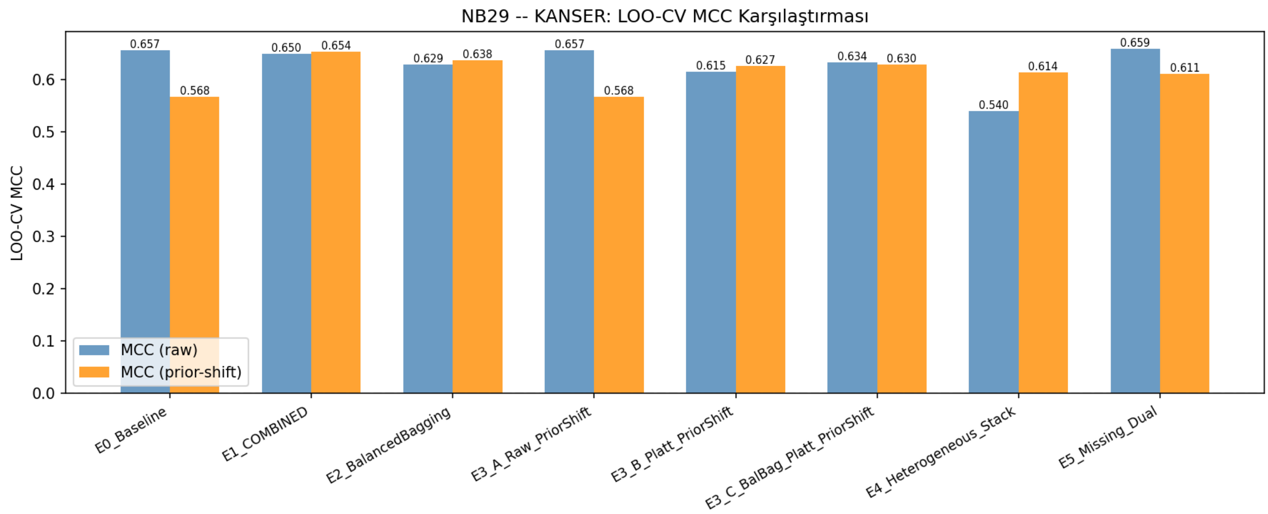
Metodoloji notu: Tum KANSER-uzeri deneyler (E0,E2,E3,E4,E5) gercek 5-fold out-of-fold (OOF) tahmin ile degerlendirildi (self-prediction leakage onlendi). E1 (COMBINED) modeli MASTER+PAH+CFTR'de egitilip KANSER'de test edildi (cross-panel, leakage yok).

1. Deney Karsilastirmasi (LOO-MCC prior sirali)

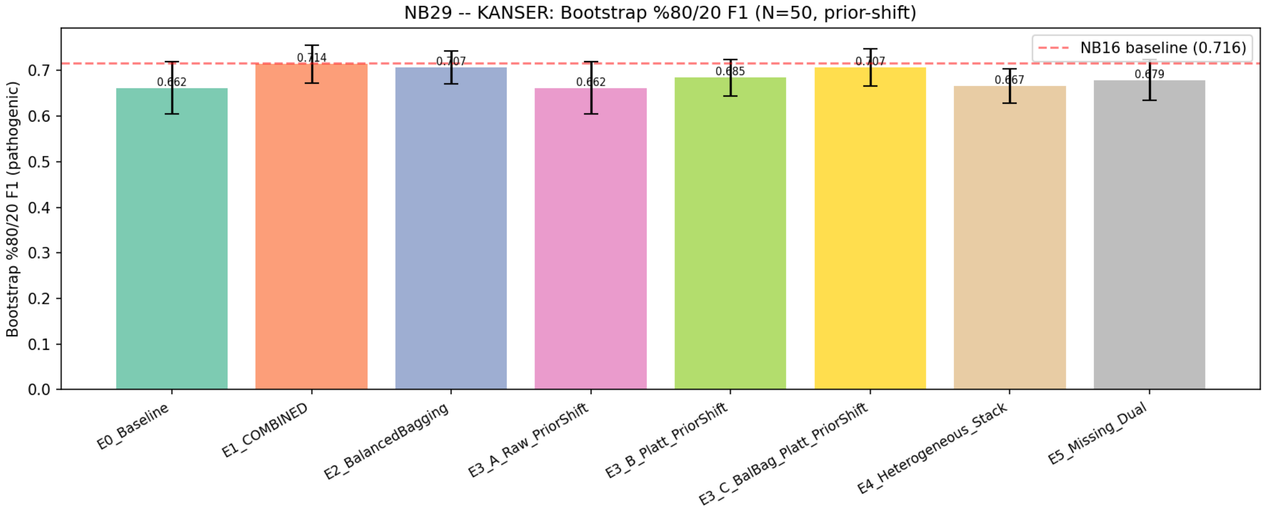
Deney	n_tr	MCC(raw)	MCC(pri)	Boot-F1	Boot-std	AUPRC	Prec	Recall	FP	FN
E1_COMBINED	3414	0.650	0.654	0.714	0.041	0.964	0.943	0.804	13	52
E2_BalancedBagging	385	0.629	0.638	0.707	0.035	0.940	0.941	0.789	13	56
E3_C_BalBag_Platt_PriorS	385	0.634	0.630	0.707	0.041	0.936	0.941	0.781	13	58
E3_B_Platt_PriorShift	385	0.615	0.627	0.684	0.040	0.919	0.933	0.792	15	55
E4_Heterogeneous_Stack	385	0.540	0.614	0.667	0.038	0.930	0.922	0.800	18	53
E5_Missing_Dual	385	0.659	0.611	0.679	0.045	0.929	0.932	0.777	15	59
E0_Baseline	385	0.657	0.568	0.662	0.058	0.929	0.936	0.721	13	74
E3_A_Raw_PriorShift	385	0.657	0.568	0.662	0.058	0.929	0.936	0.721	13	74

Birincil metrik: Bootstrap %80/20 pathogenic-F1 (N=50, final benign-agirlikli dagilim). AUPRC literatur birincil metrigidir (Lee et al. 2023).

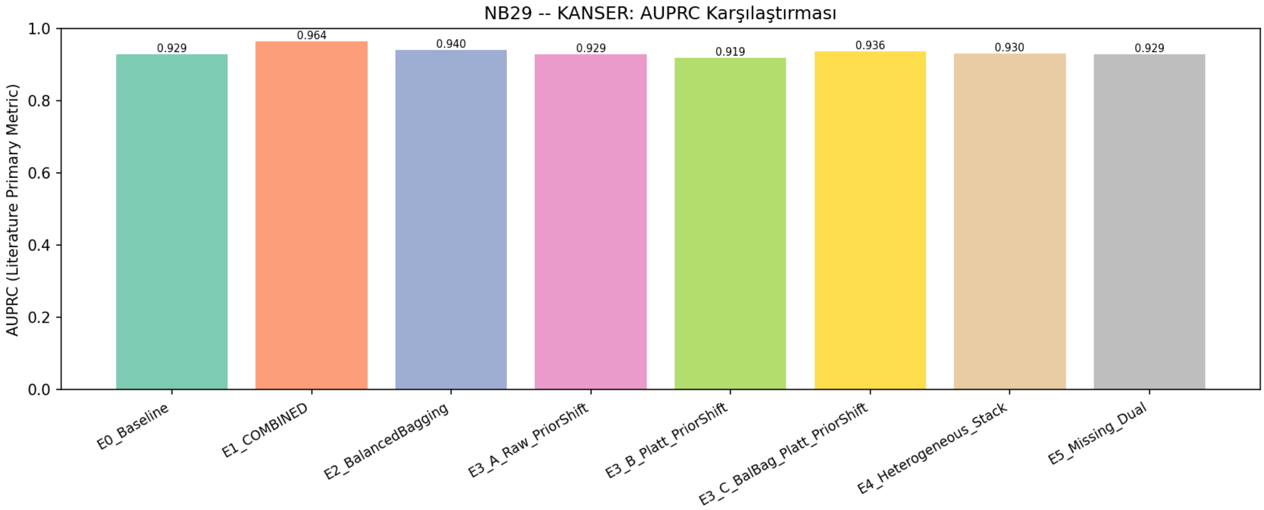
2. LOO-CV MCC (raw vs prior-shift)



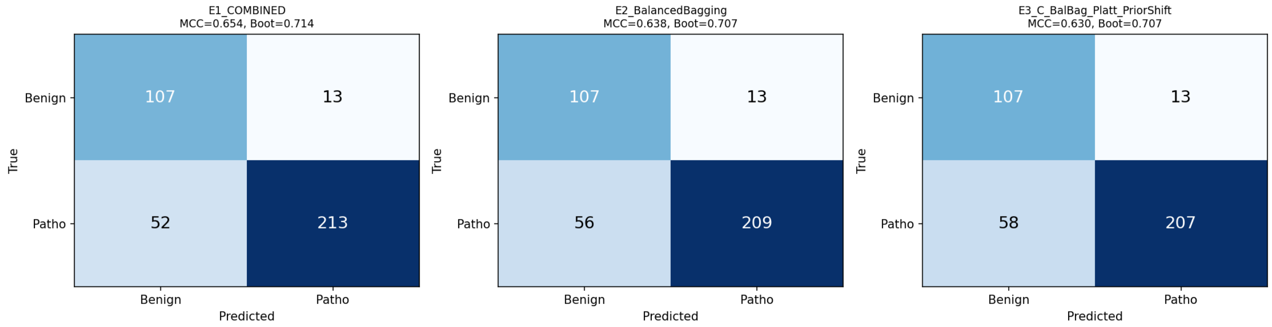
3. Bootstrap %80/20 F1



4. AUPRC (Literatur Birincil Metrigi)



5. Confusion Matrix (En Iyi 3 Deney)



6. Prior-shift Etkisi (Saerens 2002)

Deney	MCC(raw)	MCC(prior)	delta
E0_Baseline	0.6571	0.5683	-0.0887
E1_COMBINED	0.6496	0.6542	+0.0046
E2_BalancedBagging	0.6291	0.6378	+0.0087
E3_A_Raw_PriorShift	0.6571	0.5683	-0.0887
E3_B_Platt_PriorShift	0.6151	0.6273	+0.0122
E3_C_BalBag_Platt_PriorShift	0.6337	0.6297	-0.0040
E4_Heterogeneous_Stack	0.5401	0.6139	+0.0738
E5_Missing_Dual	0.6594	0.6111	-0.0484

Prior-shift, %69 patho egitim dagilimindan %20 patho test dagilimina post-hoc adapte eder. Calibrate-then-shift (Alexandari 2020): kalibrasyon ONCE, prior-shift SONRA -> kalibresiz EM'in zararli etkisi onlenir (NB21/NB22 PAH bulgusu).

7. Overfit Kontrolu (train F1 vs OOF F1)

Deney	Train-F1	OOF-F1	Gap	Durum
E0_Baseline	1.0000	0.8145	0.1855	ORTA
E1_COMBINED	0.9822	0.8676	0.1146	OK
E2_BalancedBagging	0.9027	0.8583	0.0444	OK
E3_A_Raw_PriorShift	1.0000	0.8145	0.1855	ORTA
E3_B_Platt_PriorShift	0.9648	0.8571	0.1077	OK
E3_C_BalBag_Platt_PriorShift	0.9206	0.8536	0.0670	OK
E4_Heterogeneous_Stack	0.7983	0.8566	-0.0583	OK
E5_Missing_Dual	1.0000	0.8477	0.1523	ORTA

8. Tartisma ve Literatur Temeli

KANSER paneli (n=388, pos=268/%69, neg=120) en dengeli alt-settir; n=120 benign CFTR'nin 5x kati -> bootstrap CI'lar dar ve istatistiksel olarak guvenilir (NB16 std=0.033).

- COMBINED pooling (E1): BioData Mining 2026 -- gene-specific veri kit oldugunda disease/COMBINED pooling kazanir (n=388 bu kategoride).
- BalancedBagging (E2): Chatterji 2022 NeurIPS minimax-optimal; overfit gap'i azaltir.
- Calibrate-then-shift (E3): Alexandari 2020 ICML -- kalibrasyon sonrasi prior-shift.
- Heterogeneous stack (E4): CTpredX 2025 + Lee 2023 -- in-silico skorlar ham feature, RF base, LR meta (GBM meta overfit eder).
- Missing-dual (E5): KANSER'e ozgu -- missing label-sinyali farki %35.6 (PAH'ta %0.4); AL_16..AL_25 leakage riski icin flag'li/flagsiz cift model.

NB16 referans: KANSER stack_lr %80/20 F1=0.716, CI=[0.67-0.77]. En iyi NB29 deneyi: E1_COMBINED (Boot=0.714). NB16 referansi gecilemedi.

NB16 AUPRC olculmedi; AUPRC mutlak deger olarak raporlanir (uydurma baseline kullanilmaz).